

Isaac Mubiru

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PROFESSIONAL SUMMARY

Senior backend engineer with 6+ years building **real-time, network-driven systems in Node.js** — live two-way data synchronization, gRPC services, USSD gateways, and push infrastructure. Led the end-to-end, zero-downtime migration of 20+ production applications from Parse/MongoDB to Firebase/GCP, reporting directly to the CTO: a multi-month project designed, built, verified, and cut over single-handedly. Comfortable across Python, Linux, Docker/Kubernetes, and GCP/AWS. Six years of fully remote, async-first work with US and Singapore teams — a 1-hour offset to South Africa would be the smallest of my career.

CORE COMPETENCIES

Node.js

Real-Time Data Sync

Network-Driven Systems (gRPC · Pub/Sub · USSD)

Python

Linux & Shell

Docker & Kubernetes

GCP & AWS

React · HTML · CSS

WORK EXPERIENCE

MTailor

Sep 2022 – Present

Software Engineer — US Remote · Reporting directly to the CTO

- Built **real-time two-way sync between MongoDB and Firestore** using Node.js and Google Pub/Sub — resolving concurrent writes across two live datastores under production traffic
- Led the end-to-end migration of **20+ applications** to Firebase/GCP with **zero downtime**, owning design, implementation, data verification, and cutover
- Built a 3D visualisation feature using **video overlay and ffmpeg media pipelines**, increasing buyer conversion
- Saved **\$5,000/month** by migrating all services off AWS; moved file storage from S3 to GCS via Python tooling
- Shipped Express Shipping end to end, generating an additional \$40 revenue per order

MTailor

Mar 2022 – Jul 2022

Backend Contractor — US Remote

- Built a migration pipeline (MongoDB → Firestore) with complex transformation logic and data-parity verification before cutover
- Wrote engineering documentation for Node.js and Python SDKs; trained the Ops team on backup procedures

Dr Wealth

Aug 2021 – Jan 2022

Software Engineer — Singapore Remote

- Led optimisation of a PWA delivering **real-time stock market prices**; built jobs querying 2M+ Firestore records against Morningstar APIs to keep customer prices current
- Extended backend APIs with Express.js and Firebase Cloud Functions; built responsive UI pages in React and Tailwind

CodeBits

Jan 2020 – Jul 2021

Software Engineer / Team Lead — Uganda Remote

- Built a **USSD service** for legal aid providers communicating with a Node.js backend via **gRPC** — session-based real-time flows over a stateless transport
- Architected a microservices backend using Apache Kafka, **Docker, and Kubernetes** for FIDA Uganda's case management platform
- Led a team of 4 developers; shipped a cross-platform React Native app with **Firebase push notifications** for LASPNET

Mind2matter.co

Jan 2022 – Jul 2022

Software Engineer — US Remote

- Built backends for DeFi applications using Web3 and Node.js; delivered React UIs from Figma under tight timelines

PROJECTS

Real-Time Pub/Sub Sync Pipeline

Live MongoDB ↔ Firestore two-way sync supporting parallel production traffic — concurrent-write conflict handling, ordering, and idempotent consumers during a zero-downtime migration.

Node.js · Google Pub/Sub · MongoDB · Firestore

Healthcare WhatsApp Chatbot

Real-time conversational service integrating OpenAI and Google Vision for prescription OCR via the WhatsApp Business API.

Node.js · OpenAI · Google Vision · WhatsApp Business API

Marketplace Recommendation Engine

Collaborative and content-based filtering engine built on MongoDB for service-provider matching.

Node.js · MongoDB

EDUCATION

BSc Computer Engineering — Makerere University

2018 – 2022

Kampala, Uganda

SKILLS

Proficient: Node.js, Python, Firebase, GCP, MongoDB, SQL, React, HTML, CSS, Git, Linux **Intermediate:** Go, AWS, Rust, React Native

Infrastructure: Docker, Kubernetes, Kafka, Pub/Sub, NATS Streaming, gRPC, Heroku **Languages:** English (fluent), Kiswahili